Simulating Age-Period-Cohort Data

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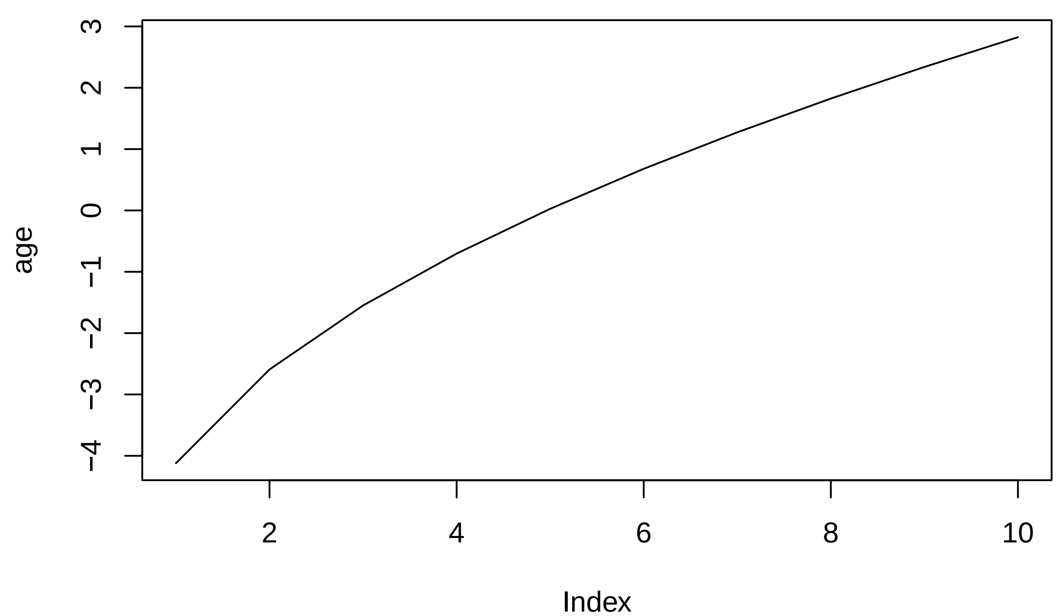
This vignette simulates age-period-cohort data from known, hand-specified age, period and cohort effects, fits `bamp()` to the simulated cases, and checks that the fitted effects recover the true ones – a good way to sanity check the method, or to plan how much data a study design would need. For a first introduction to `bamp()`, see *vignette("bamp", package="bamp")*; for other model specifications, see *vignette("modeling", package="bamp")*.

Specifying the true effects

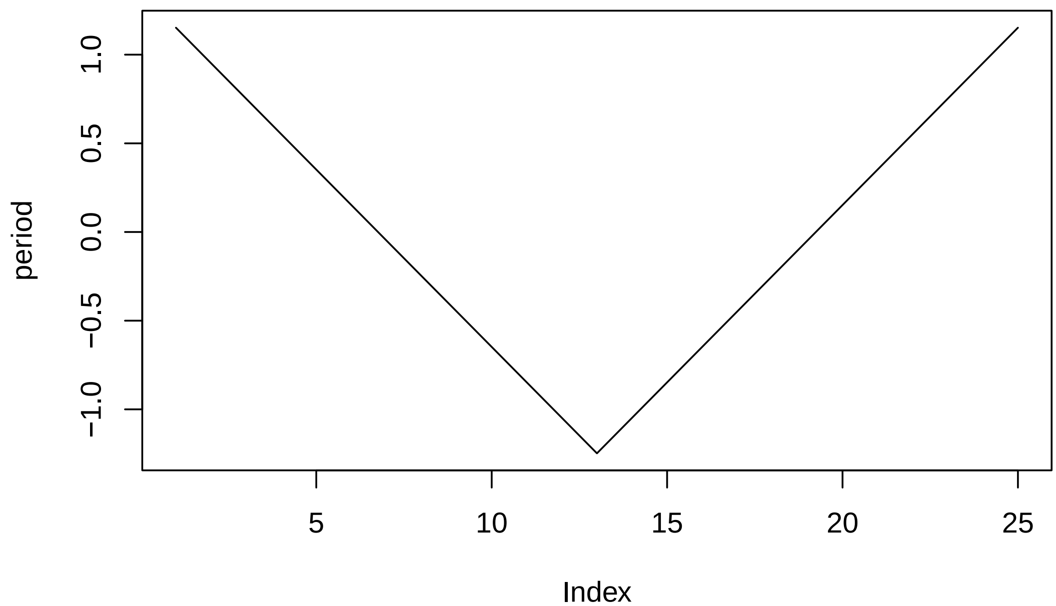
We first fix the three true effects on the log-odds scale; each is centred (mean zero) since only their shape, not their level, is identified (see *vignette("modeling", package="bamp")*). The age effect is an increasing, concave curve:

```
age=2*sqrt(seq(1,20,length=10))
age<- age-mean(age)
plot(age, type="l")
```



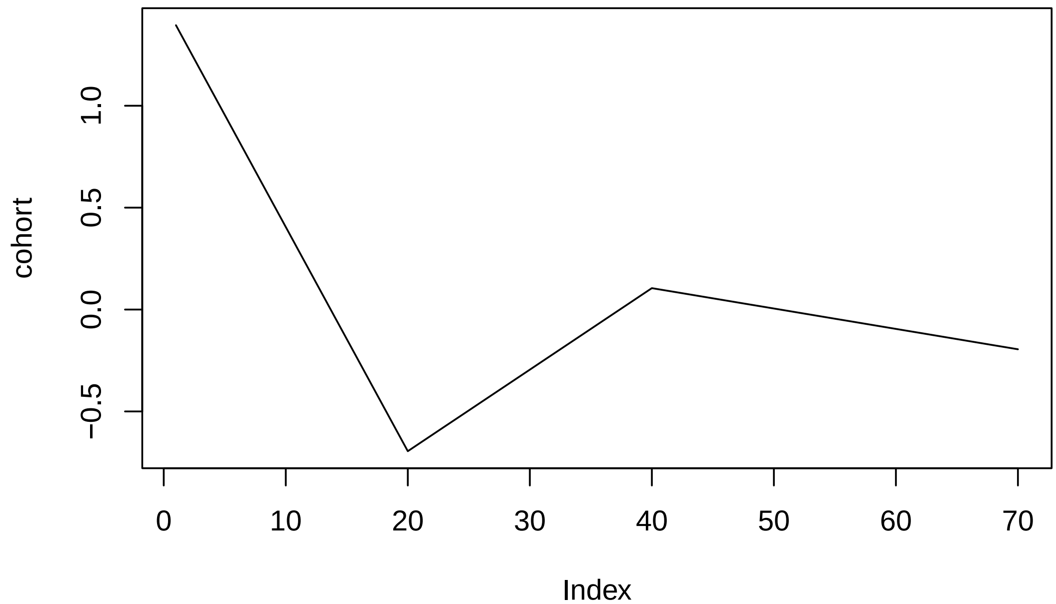
The period effect is V-shaped: it falls for the first half of the observed periods, then rises again over the second half:

```r
period=25:1
period[13:25]<-13:25
period<-period/5
period<-period-mean(period)
plot(period, type="l")
```



The cohort effect falls for the earliest cohorts, rises through the middle ones, then slightly falls for the most recent cohorts. With `periods_per_agegroup = 5` and 10 age groups spanning 25 periods, there are `periods_per_agegroup * (10 - 1) + 25 = 70` cohorts:

```r
periods_per_agegroup=5
number_of_cohorts <- periods_per_agegroup*(10-1)+25
cohort<-rep(0,70)
cohort[1:20]<-(19:0)
cohort[21:40]<- (1:20)/2
cohort[41:70]<- 10
cohort=cohort-(1:70)/10
cohort<-cohort/10
cohort<-cohort-mean(cohort)
plot(cohort, type="l")
```



Simulating data from the true effects

`apcSimulate()` combines an intercept (here `-10`, on the log-odds scale) with the three true effects and a population of `1 000 000` per Lexis cell to draw simulated case counts, returning `cases` and `population` matrices just like the bundled `apc` data example:

```r
simdata<-apcSimulate(-10, age, period, cohort, periods_per_agegroup, 1e6)
print(simdata$cases)
```

```
##       [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10]
##  [1,]    1   16   32   54   88  131  383 1131 3358  9599
##  [2,]    2    9   22   31   77  119  272  843 2531  7097
##  [3,]    1    6   21   46   63   92  226  613 1942  5257
##  [4,]    2    4   11   31   52   89  162  492 1330  3866
##  [5,]    1    2   13   28   48   78  130  333  999  2689
##  [6,]    0    2    5   20   35   61  107  257  745  2084
##  [7,]    0    6    8   20   32   60   75  188  542  1524
##  [8,]    0    1    7   16   34   54   80  134  440  1161
##  [9,]    2    3   10   14   34   43   67  111  282   867
## [10,]    0    1    2   14   29   43   58   70  221   604
## [11,]    0    0    8   11   12   35   44   63  173   436
## [12,]    0    0    3   10   25   23   38   56  111   327
## [13,]    1    1    5    2   17   19   32   44   76   251
## [14,]    0    2    6   12   22   22   34   67   88   283
## [15,]    1    1    3   15   21   30   53   70   98   291
## [16,]    1    1    3    9   20   37   70   93  125   332
## [17,]    0    4   10   20   36   67   71  104  153   346
## [18,]    0    3    4   26   38   75   98  157  178   378
## [19,]    0    3    8   20   44   92  135  196  259   393
```

```
## [20,]    0    3   19   26   41  106  155  275  344   406
## [21,]    1    8   18   31   66  161  216  297  438   559
## [22,]    1    4   17   37   79  161  292  343  590   696
## [23,]    3    3   16   43  102  190  354  486  657   909
## [24,]    3    7   27   45  111  275  466  698  864  1168
## [25,]    0   11   20   73  166  305  587  803 1095  1441
```

Recovering the effects with `bamp()`

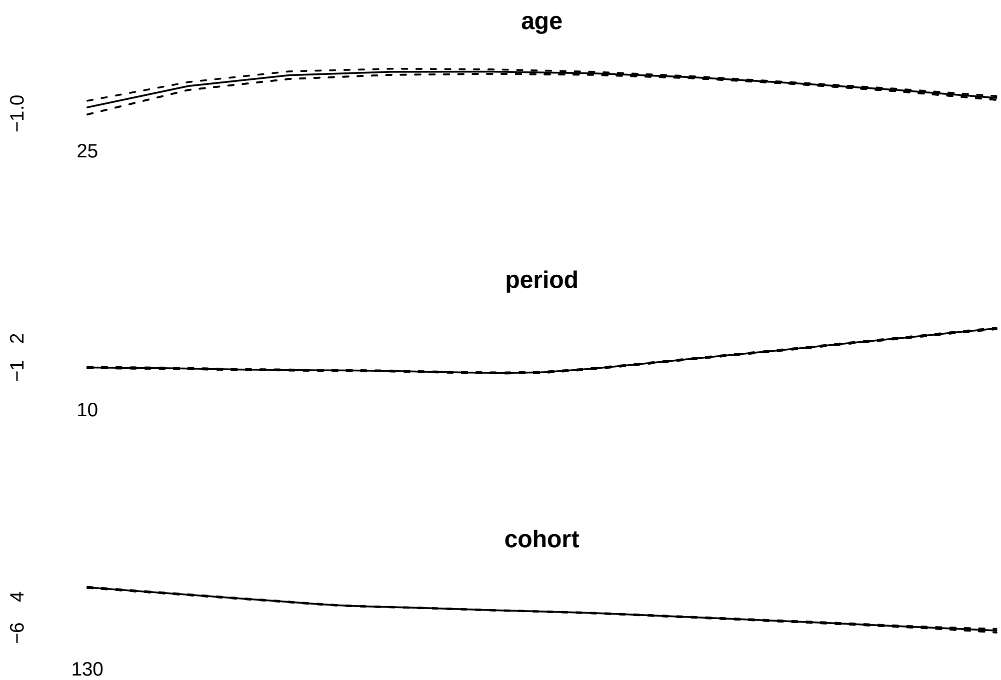
We fit the same first-order random-walk model used throughout the other vignettes to the simulated data:

```r
simmod <- bamp(cases = simdata$cases, population = simdata$population, age = "rw2",
period = "rw2", cohort = "rw2", periods_per_agegroup =periods_per_agegroup)
```

```r
print(simmod)
```

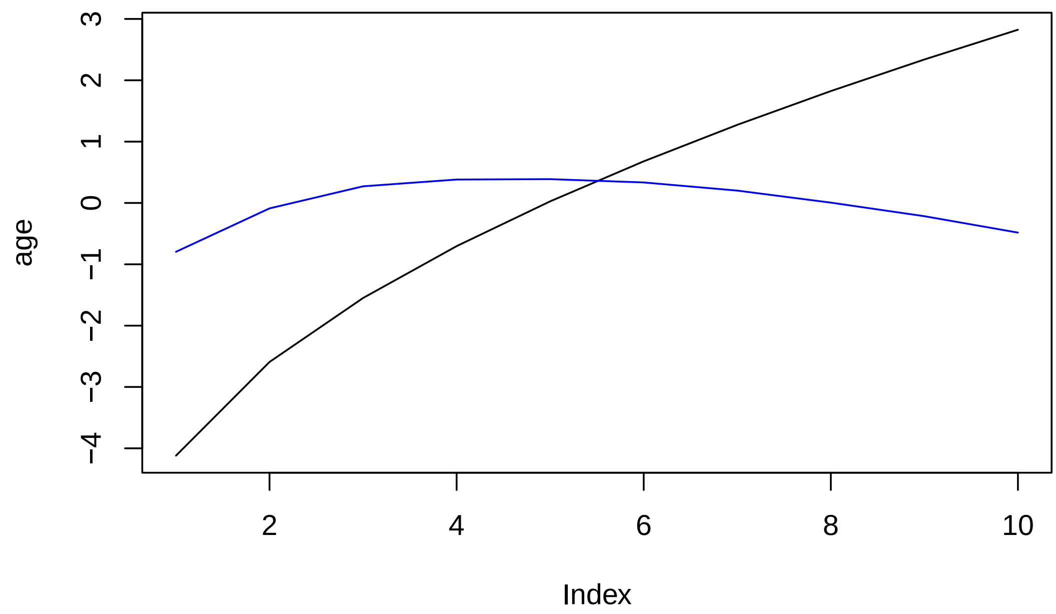
```
##
##  Model:
## age (rw2)  - period (rw2)  - cohort (rw2) model
## Deviance:     263.19
## pD:            41.09
## DIC:          304.28
##
##
##  Hyper parameters:                    5%          50%          95%
## age                               2.958        7.207       14.339
## period                          132.437      257.810      459.996
## cohort                          978.719     1907.714     3368.035
##
##
## Markov Chains convergence checked succesfully using Gelman's R (potential scale reduction factor).
```

```r
par(mfrow=c(3,1))
plot(simmod)
```

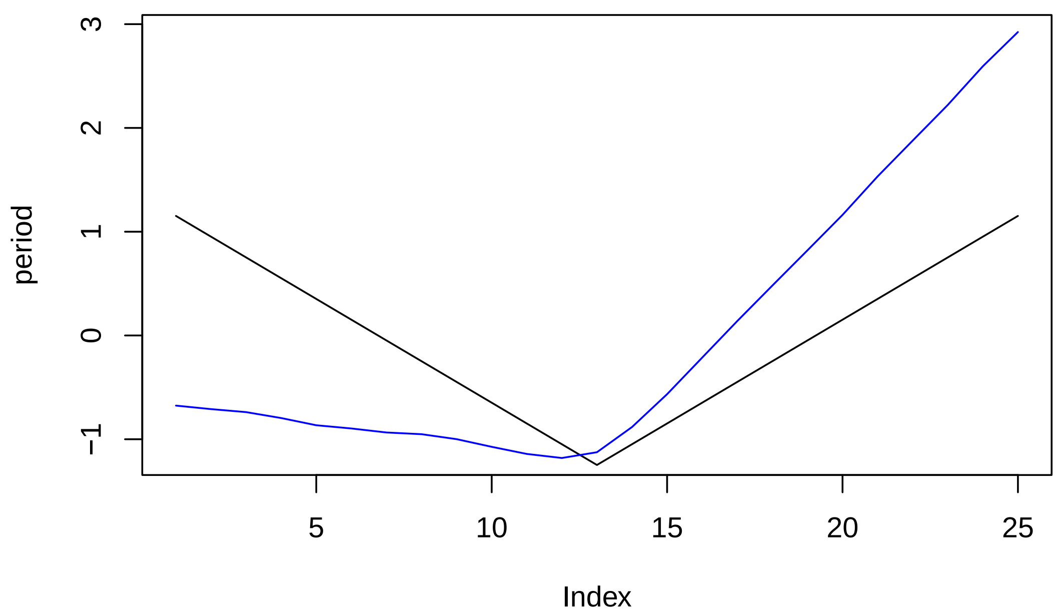


Since we know the true effects here (unlike with real data), we can check directly how well `bamp()` recovered them: the black line is the true effect, the blue line the fitted posterior median (`effects()`). Because the age, period and cohort effects are only identified up to a shared linear trend (see *vignette("modeling", package="bamp")*), the recovered curves show the same nonlinear (curvature) features as the truth, but are linearly shifted relative to it.

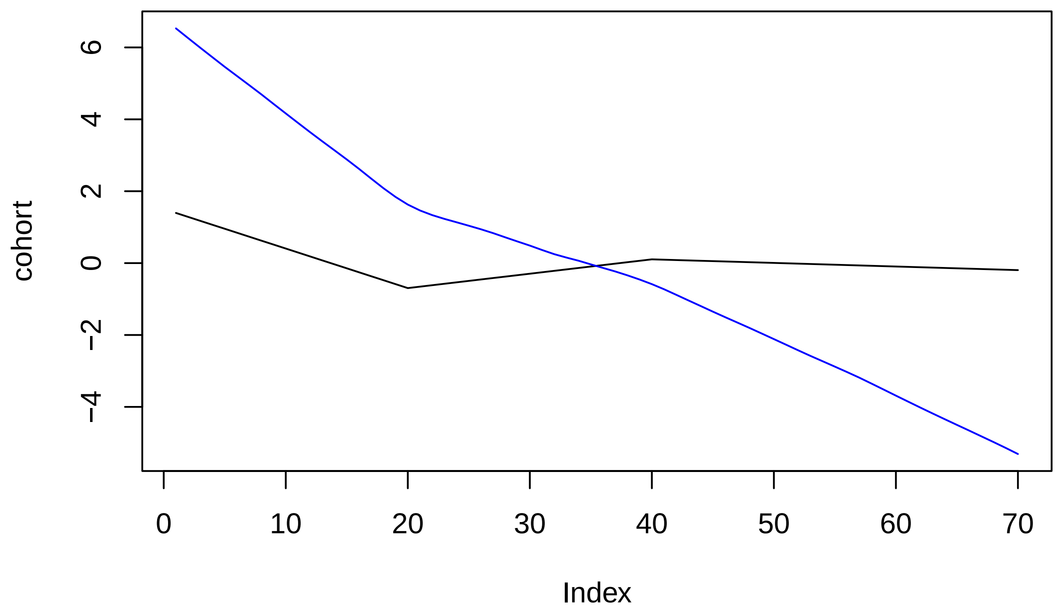
```
effects<-effects(simmod)
plot(age, type="l")
lines(effects$age, col="blue")
```




```r
plot(period, type="l", ylim=range(effects$period))
lines(effects$period, col="blue")
```



```r
plot(cohort, type="l", ylim=range(effects$cohort))
lines(effects$cohort, col="blue")
```



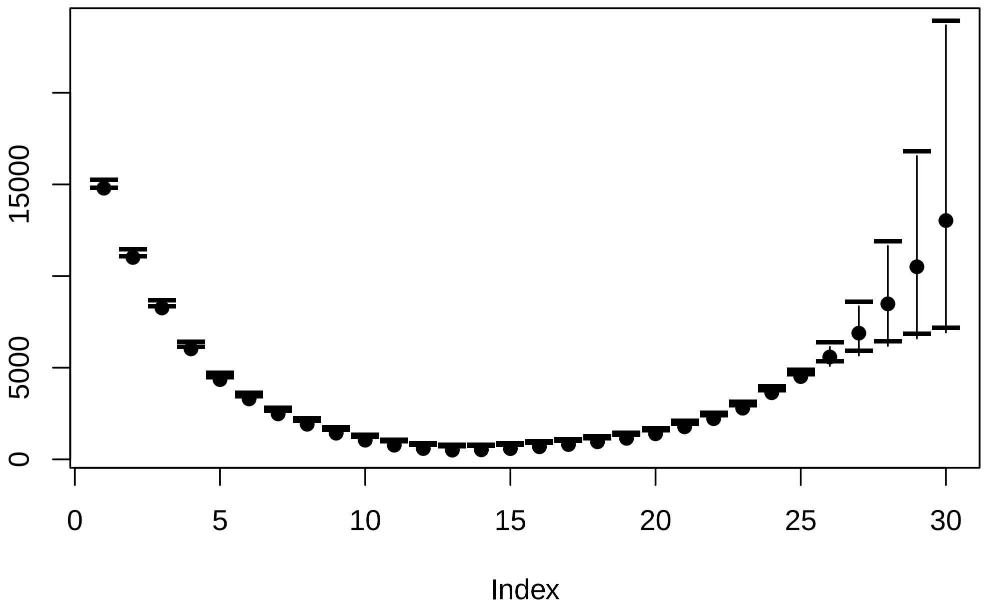
Prediction

As in *vignette("prediction", package="bamp")*, the fitted random walks can be extrapolated to forecast cases for periods beyond the simulated data – here 5 more, so the supplied `population` array covers 30 periods (25 observed + 5 forecast) across the 10 age groups:

```r
prediction<-predict_apc(simmod, periods=5, population=array(1e6,c(30,10)), quantiles=c(.1,.5,.9))
```

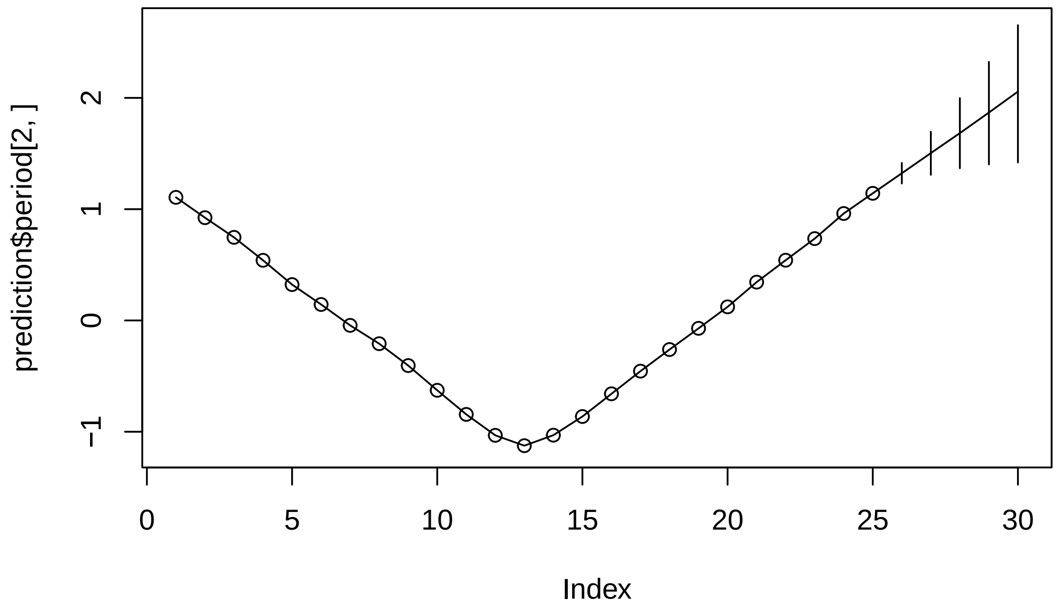
Total predicted cases per period are shown with their median (solid points) and 80% credible interval (dashes):

```r
plot(prediction$cases_period[2,], ylim=range(prediction$cases_period),ylab="",pch=19)
points(prediction$cases_period[1,],pch="-",cex=2)
points(prediction$cases_period[3,],pch="-",cex=2)
for (i in 1:30)lines(rep(i,3),prediction$cases_period[,i])
```



The period effect itself extends the same way into the 5 forecast periods:

```
plot(prediction$period[2,], type="l", ylim=range(prediction$period))
points(prediction$period[2,1:25])
for (i in 26:30)lines(rep(i,3),prediction$period[,i])
```



Period covariate

Finally, we refit the simulated data with a period covariate scaling the period effect (see *vignette("modeling", package="bamp")*). A period covariate is a positive multiplier, so it must be larger than zero.

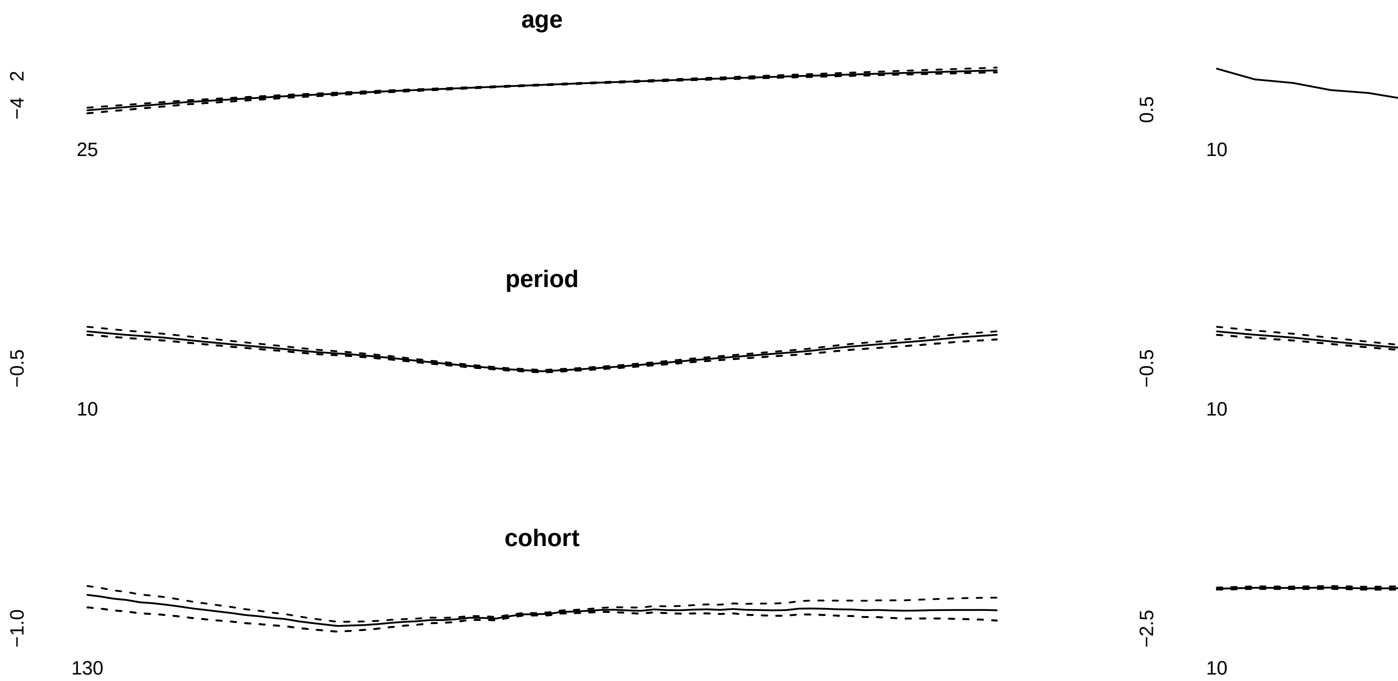
```
cov_p<-exp(rnorm(30,period,.1))
```

```
simmod2 <- bamp(cases = simdata$cases, population = simdata$population, age = "rw1",
period = "rw1", cohort = "rw1", periods_per_agegroup = periods_per_agegroup,
period_covariate = cov_p)
```

```
print(simmod2)
```

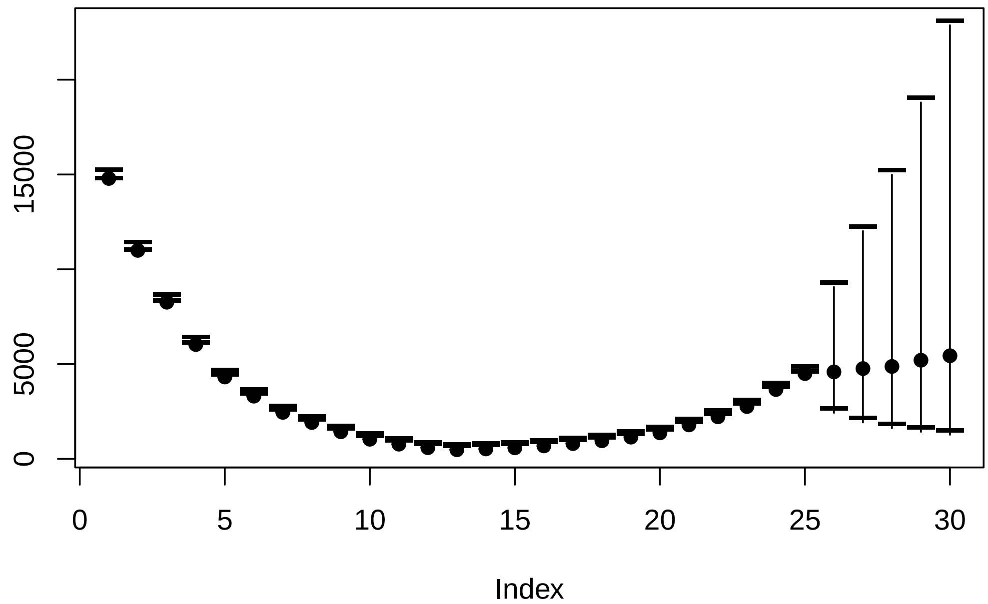
```
##
##  Model:
## age (rw1)  - period (rw1)  - cohort (rw1) model
## Deviance:     254.82
## pD:            69.69
## DIC:          324.52
##
##
##  Hyper parameters:                   5%         50%         95%
## age                               0.587       1.366       2.666
## period                            2.299       4.067       6.648
## cohort                           74.653     110.732     160.572
##
##
## Markov Chains convergence checked succesfully using Gelman's R (potential scale reduction factor).
```

```
par(mfrow=c(3,1))
plot(simmod2)
```




```
prediction2<-predict_apc(simmod2, periods=5, population=array(1e6,c(30,10)), quantiles=c(.1,.5,.9))
```

```
plot(prediction2$cases_period[2,], ylim=range(prediction2$cases_period),ylab="",pch=19)
points(prediction2$cases_period[1,],pch="-",cex=2)
points(prediction2$cases_period[3,],pch="-",cex=2)
for (i in 1:30)lines(rep(i,3),prediction2$cases_period[,i])
```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5000
15000
0
5
10
15
20
25
30
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